
Iacovos G. Kolokasis

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I am a PhD candidate in Computer Science at the University of Crete and a Graduate Research Assistant at the Institute of Computer Science, Foundation for Research and Technology – Hellas, specializing in managed runtimes, garbage collection, memory management, and storage systems. My doctoral work has focused mainly on OpenJDK HotSpot and G1GC, including heap resizing, memory partitioning, and tiered heap management for data-intensive Java applications. My background also includes interactions between the JVM interpreter and JIT compilers, JVM performance profiling, NVMe SSDs, the Linux memory-mapped I/O path, Apache Spark, Flink, Elasticsearch, Cassandra, Neo4j, and Lucene.

TECHNICAL SKILLS

Programming Languages: C/C++, Java, Scala, Python, Bash, Awk

Performance Profiling & Debugging: async-profiler, Linux perf, FlameGraphs, GDB, JDB, JVM GC logs

Build & Infrastructure: CMake, Make, Git, Containers, Linux

Data-Intensive Systems & Workloads: Apache Spark, Apache Flink, Elasticsearch, Lucene, Cassandra, Neo4j, Giraph

Managed Runtimes & Garbage Collection: OpenJDK/HotSpot, G1GC, Parallel Scavenge, Interpreter, C1/C2 compilers

PROFESSIONAL EXPERIENCE

Graduate Research Assistant at [CARV, ICS-FORTH](#)

PhD Projects:

- Designed and implemented **TeraHeap**, an OpenJDK/HotSpot tiered-heap mechanism that extends the JVM heap over NVMe SSDs to reduce DRAM pressure in data-intensive Java applications; initially implemented for Parallel Scavenge and later ported to the actively developed G1GC version.
- Designed and implemented **FlexHeap**, an extension to OpenJDK 21's G1GC heap-resizing policy that dynamically repartitions DRAM between the JVM heap and the OS page cache based on garbage-collection cost and mutator I/O stalls.
- Developed **SmartSweep**, a garbage-collection mechanism for tiered managed heaps in disaggregated-memory environments that reclaims remote-heap space using approximate region-level liveness, avoiding expensive scans of remote memory.

Advisors: Prof. Angelos Bilas and Prof. Polyvios Pratikakis

Nov. 2017 -
Now

Research intern at [Huawei Research Center, Munich, Germany](#)

- Designed, developed, and evaluated an extension to the Parallel Scavenge garbage collector in OpenJDK for disaggregated memory environments.

Jun. 2022 –
Sep. 2022

Research intern at [SAP, Walldorf, Germany](#)

- Improved the internal representation of the SQL parser's syntax tree to support more efficient query parsing and analysis.

Jun. 2019 –
Sep. 2019

Undergraduate Research Assistant at [CARV, ICS-FORTH](#)

- Designed and developed techniques for partitioning large-scale graphs in Apache Spark GraphX.
- Evaluated graph-partitioning strategies for Apache Spark GraphX, studying their impact on locality, communication, and performance across graph workloads.

Advisor: Prof. Polyvios Pratikakis

Jan 2017 –
Aug. 2017

EDUCATION

PhD in Computer Science at the Computer Science Department, University of Crete, Heraklion, Greece	Jan. 2021 – Now
MSc in Computer Science at the Computer Science Department, University of Crete, Heraklion, Greece	Sep. 2017 – Oct. 2020
BSc in Computer Science at the Computer Science Department, University of Crete, Heraklion, Greece	Sep. 2011 – Jul. 2017

SELECTED PUBLICATIONS

Full Papers / Journals

FlexHeap: Dynamic I/O-Aware Heap Resizing for Managed Applications. (PLDI '26) Technologies used: OpenJDK HotSpot, G1GC, eBPF, Elasticsearch, Apache Spark	2026
MaTSa: Race Detection in Java. (VMIL '25) Technologies used: OpenJDK HotSpot, Interpreter, C1/C2 compilers, G1GC	2025
SmartSweep: Efficient Space Reclamation in Tiered Managed Heaps. (MPLR '25) Technologies used: OpenJDK HotSpot, G1GC, NVMe-oF, Apache Spark, Neo4j	2025
TeraHeap: Exploiting Flash Storage for Mitigating DRAM Pressure in Managed Big Data Frameworks. (TOPLAS). Technologies used: OpenJDK HotSpot, Parallel Scavenge, NVMe SSDs, NVM, Apache Spark, Apache Giraph, Neo4j	2024
TeraHeap: Reducing Memory Pressure in Managed Big Data Frameworks. (ASPLOS '23) Technologies used: OpenJDK HotSpot, Parallel Scavenge, NVMe SSDs, NVM, Apache Spark, Apache Giraph	2023
Say Goodbye to Off-heap Caches! On-heap Caches Using Memory-Mapped I/O. (HotStorage '20) (Best Presentation Award) Technologies used: OpenJDK HotSpot, Parallel Scavenge, NVMe SSDs, Apache Spark	2020
Cut to Fit: Tailoring the Partitioning to the Computation. (GRADES-NDA) Technologies used: Apache Spark	2019

Posters

DynaHeap: Dynamic Division of DRAM between Heterogeneous Managed Heaps. (EuroSys '24)	2024
DynaHeap: Dynamic Division of DRAM between Heterogeneous Managed Heaps. (SOSP '23)	2023
TeraCache: Efficient Caching over Fast Storage Devices. MSR Cambridge PhD Virtual Workshop on Next-Generation Cloud Infrastructure'21.	2021
Garbage Collection or Serialization? Between a Rock and a Hard Place! (ACACES)	2020

FELLOWSHIPS and AWARDS

HotOS'25 Student scholarship	2025
Meta Research PhD fellowship	2022–2024
Cyprus State Scholarship Foundation PhD fellowship	2021–2024
Independent Power Transmission Operator (ADMIE) fellowship	2020,2021, 2024
HiPEAC paper award	2024
Best presentation award, HotStorage'20	2020
Stelios Orphanoudakis Fellowship	2012
Selected conference scholarships: HotOS'25, ASPLOS 2023, SOSP 2023, ASPLOS 2022, OSDI 2021, FAST 2021	

SELECTED PROFESSIONAL SERVICE & TALKS

Reviewer, ACM Transactions on Architecture and Code Optimization (TACO)	2026
Web Chair, ASPLOS 2027 Organizing Committee	2027
Program Committee Member: MICRO'26, TDIS'25, TDIS'26, and VMIL'25	2025–2026
Artifact Evaluation Committee Member: EuroSys'25, OSDI'21; ASPLOS'20, ASPLOS'21	2020–2025
Shadow Program Committee Member: EuroSys'25, EuroSys'22	2022,2025
Presentation of TeraCache: Efficient Caching over Fast Storage Devices, MSR Cambridge PhD Virtual Workshop on Next-Generation Cloud Infrastructure (invited talk)	2021
Presentation of TeraCache: Efficient Caching over Fast Storage Devices, Spark + AI Summit Europe	2020

TEACHING EXPERIENCE AS ASSISTANT

Parallel programming (CS342)	2018,2019, 2025–2026
Introduction to computer science (CS100)	2018 – 2026
Programming lab (CS255)	2021 – 2024
Data structures (CS240)	2018, 2020
Digital design (CS120)	2017